

Problem-Solution Fit

Date	30 June 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

1. CUSTOMER SEGMENT(S) [CS] Banks and financial institutions Credit officers and loan approval teams Financial analysts evaluating loan applicants Loan applicants (salaried and self-employed)	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] Limited IT budgets in smaller institutions Dependence on legacy banking systems Variable data quality from applicants Limited staff time during high-volume periods	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Manual credit review - thorough but slow and inconsistent Rule-based scoring systems - fast but rigid and less accurate Generic ML credit tools - accurate but costly and complex to integrate
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] Slow, manual loan evaluation process Inconsistent approval decisions across officers Rising non-performing assets (NPAs) Occurs with every loan application, daily	9. PROBLEM ROOT / CAUSE [RC] No automated, data-driven tool to assess creditworthiness Reliance on manual judgment leads to slow, error-prone, and inconsistent decisions	7. BEHAVIOR + ITS INTENSITY [BE] Officers manually review each applicant's documents Repetitive data entry and cross-checking credit history High intensity during peak application periods
3. TRIGGERS TO ACT [TR] New loan application submitted High volume of applications during peak periods Pressure to reduce NPAs and improve turnaround time	10. YOUR SOLUTION [SL] Smart Lender: an ML-powered web app predicting loan applicant creditworthiness Evaluates Decision Tree, Random Forest, KNN, and XGBoost models Best model (XGBoost: 94.7% train / 81.1% test accuracy) deployed via Flask Provides real-time approval predictions; designed for deployment on IBM Cloud	8. CHANNELS OF BEHAVIOR [CH] ONLINE Online: Web application, bank online loan portals
4. EMOTIONS BEFORE / AFTER [EM] Before: stress, uncertainty, time pressure After: confidence, relief, faster and fairer decisions		OFFLINE Offline: Bank branch loan desks, in-person document submission